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Subject:AIML

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Branch:electronics and computer

Q1.(Setup and Libraries)

```
import streamlit as st
```

```
import pandas as pd
```

```
import joblib
```

Q2.(Loading Model and Preprocessing Objects,

```
model = joblib.load("LR_model.pkl")
```

```
scaler = joblib.load("scaler.pkl")
```

```
encoded_columns = joblib.load("columns.pkl")
```

Q3.(Page Configuration)

```
st.set_page_config(
```

```
    page_title="Ford Car Price Predictor",
```

```
    layout="centered"
```

```
)
```

Q4.(Title and Description)

```
st.title("🚗 Ford Car Price Predictor")
```

```
st.write("Enter the car details below to predict its selling price.")
```

Q5.(Numerical Input Fields)

```
year = st.number_input(  
    "Manufacturing Year",  
    min_value=1990,  
    max_value=2025,  
    value=2018
```

```
)
```

```
mileage = st.number_input(  
    "Mileage",  
    min_value=0,  
    max_value=300000,  
    value=30000
```

```
)
```

```
tax = st.number_input(  
    "Road Tax (£)",  
    min_value=0,  
    max_value=600,  
    value=150
```

```
)
```

```
mpg = st.number_input(  
    "MPG",  
    min_value=0.0,
```

```
        max_value=100.0,  
        value=50.0  
    )  
    engineSize = st.number_input(  
        "Engine Size (L)",  
        min_value=0.0,  
        max_value=8.0,  
        value=1.5  
    )
```

Q6.(Categorical Input using Dropdowns)

```
transmission = st.selectbox(  
    "Transmission",  
    ["Automatic", "Manual", "Semi-Auto"]  
)  
fuelType = st.selectbox(  
    "Fuel Type",  
    ["Petrol", "Diesel", "Hybrid", "Electric", "Other"]  
)
```

Q7.(Text Input and Predict Button)

```
model_name = st.text_input("Enter Car Model Name")
```

```
predict_button = st.button("Predict Price")
```

Q8.(Creating Input DataFrame & Encoding) When the Predict button is clicked:

Create a pandas DataFrame from user inputs • Perform One-Hot Encoding using `pd.get_dummies()`. • Align the columns with the # Run prediction when the button is clicked

```
if predict_button:
```

```
    input_data = pd.DataFrame({
        "model": [model_name],
        "year": [year],
        "transmission": [transmission],
        "mileage": [mileage],
        "fuelType": [fuelType],
        "tax": [tax],
        "mpg": [mpg],
        "engineSize": [engineSize]
    })

    input_encoded = pd.get_dummies(input_data)

    input_encoded = input_encoded.reindex(columns=encoded_columns, fill_value=0)
    training_columns = encoded_columns.
```

Q9.(Feature Scaling and Prediction),# Identify the numerical columns that were scaled during training

```
numeric_columns = ["year", "mileage", "tax", "mpg", "engineSize"]
```

```
input_encoded[numeric_columns] = scaler.transform(input_encoded[numeric_columns])
```

```
predicted_price = model.predict(input_encoded)
st.success(f"Predicted Car Price: £{predicted_price[0]:,.2f}")
```

Q10.(Mini Project - Complete Streamlit App)

```
import streamlit as st
import pandas as pd
import joblib

st.set_page_config(
    page_title="Ford Car Price Predictor",
    layout="centered"
)

try:
    model = joblib.load("LR_model.pkl")
    scaler = joblib.load("scaler.pkl")
    encoded_columns = joblib.load("columns.pkl")
except Exception as e:
    st.error(f"Error loading files: {e}")
    st.stop()

st.title(" Ford Car Price Predictor")

st.write("Enter the car details below to predict its selling price.")

st.markdown("---")

model_name = st.text_input("Car Model")

year = st.number_input(
    "Manufacturing Year",
    min_value=1990,
```

```
        max_value=2025,  
        value=2018  
    )  
    mileage = st.number_input(  
        "Mileage",  
        min_value=0,  
        max_value=300000,  
        value=30000  
    )  
    tax = st.number_input(  
        "Road Tax",  
        min_value=0,  
        max_value=600,  
        value=150  
    )  
    mpg = st.number_input(  
        "MPG",  
        min_value=0.0,  
        max_value=100.0,  
        value=55.0  
    )  
    engineSize = st.number_input(  
        "Engine Size",  
        min_value=0.8,  
        max_value=6.0,
```

```
        value=1.5
    )

    transmission = st.selectbox(
        "Transmission",
        ["Automatic", "Manual", "Semi-Auto"]
    )

    fuelType = st.selectbox(
        "Fuel Type",
        ["Petrol", "Diesel", "Hybrid", "Electric", "Other"]
    )

    st.markdown("---")

    predict_button = st.button("Predict Price")

    if predict_button:
        try:
            input_data = pd.DataFrame({
                "model":[model_name],
                "year":[year],
                "transmission":[transmission],
                "mileage":[mileage],
                "fuelType":[fuelType],
                "tax":[tax],
                "mpg":[mpg],
                "engineSize":[engineSize]
            })

            input_encoded = pd.get_dummies(input_data)
```



```

input_encoded = input_encoded.reindex(
    columns=encoded_columns,
    fill_value=0
)
numeric_columns = [
    "year",
    "mileage",
    "tax",
    "mpg",
    "engineSize"
]
input_encoded[numeric_columns] = scaler.transform(
    input_encoded[numeric_columns]
)
predicted_price = model.predict(input_encoded)
st.success(
    f" Predicted Car Price: £{predicted_price[0]:.2f}"
)
except Exception as e:
    st.error(f"Prediction Error: {e}")

```

Optional Task:

Create a Frontend for Model you created in Optional Assignment 20

```
import streamlit as st

import pandas as pd

import joblib

model = joblib.load("model.pkl")

st.set_page_config(page_title="ML Model Predictor", layout="centered")

st.title("Machine Learning Prediction App")

st.write("Enter the required details below.")

feature1 = st.number_input("Feature 1", value=0.0)

feature2 = st.number_input("Feature 2", value=0.0)

feature3 = st.number_input("Feature 3", value=0.0)

if st.button("Predict"):

    input_data = pd.DataFrame({

        "Feature1": [feature1],

        "Feature2": [feature2],

        "Feature3": [feature3]

    })

    prediction = model.predict(input_data)

    st.success(f"Prediction: {prediction[0]}")
```